2da Clase 22/8/2025

75.43 Introducción a los Sistemas Distribuidos
95.60 Redes y Aplicaciones Distribuidas
TA048 Redes

Tema: Capa de Aplicación

Capítulo 2 completo de *Computer Networking : A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson Edition: 8th, 2021.*

Dr. Ing. J. Ignacio Alvarez-Hamelin

Teórica (viernes 19-22)

Dr. Ing. J. Ignacio Alvarez-Hamelin

Práctica 1 (martes 19-22)

JTP: Lic. Juan Ignacio López Pecora
Ay 1ro: Ing. Agustin Horn
Ay 2do: Hugo Huzan

Práctica 2 (jueves 18-21)

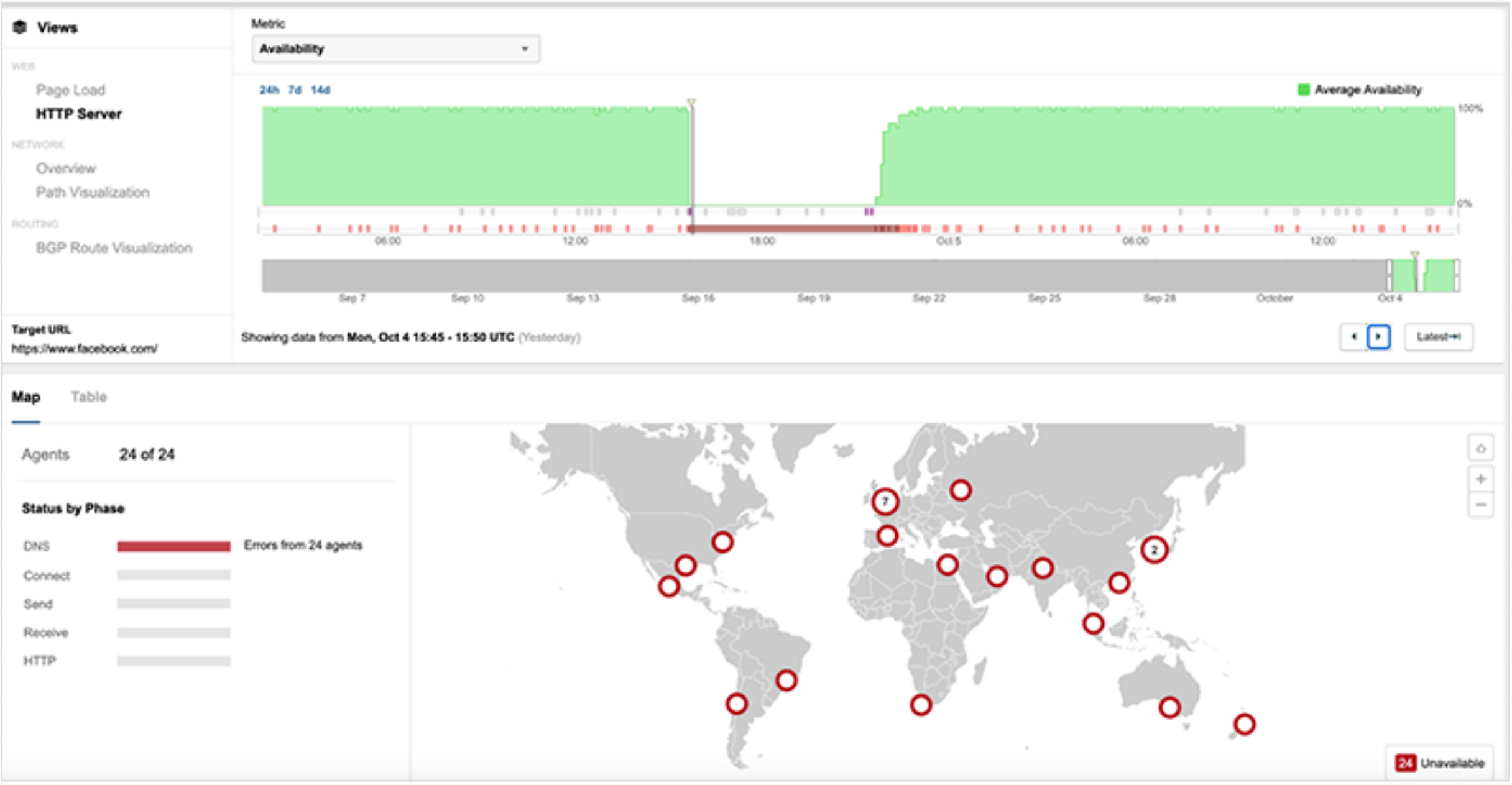
JTP: Ing. José Luis Balbiano
Ay 1ro: Mag. Javier Scodelaro
Ay 2do: Ezequiel Mundani Vegega

- Dinámica de las clases
- Condiciones de aprobación y regularidad
- Código de conducta
- Reglamento de la materia (disponible en el campus)
- Herramientas: software libre, por ejemplo:
  - ⋆ **Wireshark** https://www.wireshark.org
  - ⋆ línea de comando *UNIX: ping, traceroute, dig, ifconfig, netstat
- *Comunicación*: ***Campus*** ***+ slack + e-mails***
- Bibliografía:
  - ❖ **Computer Networking : A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson Edition: 8th, 2021.**
  - ❖ **RFCs (Request For Comments)**
  - ❖ **trabajos científicos**

#2025-2c

https://www.thousandeyes.com/blog/facebook-outage-analysis?t&utm_source=perplexity

On October 4, 2021, Facebook experienced a prolonged outage preventing users from around the globe from reaching its services.



Trace for Paris, France (Trial)

```
com.    172800  IN      NS      m.gtld-servers.net.
com.    172800  IN      NS      g.gtld-servers.net.
com.    172800  IN      NS      a.gtld-servers.net.
com.    172800  IN      NS      c.gtld-servers.net.
com.    172800  IN      NS      b.gtld-servers.net.
com.    172800  IN      NS      e.gtld-servers.net.
com.    172800  IN      NS      i.gtld-servers.net.
com.    172800  IN      NS      d.gtld-servers.net.
com.    172800  IN      NS      h.gtld-servers.net.
com.    172800  IN      NS      k.gtld-servers.net.
com.    172800  IN      NS      f.gtld-servers.net.
com.    172800  IN      NS      j.gtld-servers.net.
com.    172800  IN      NS      l.gtld-servers.net.
;; Received 490 bytes from 193.0.14.129(k.root-servers.net.) in 10 ms

facebook.com.   172800  IN      NS      a.ns.facebook.com.
facebook.com.   172800  IN      NS      b.ns.facebook.com.
facebook.com.   172800  IN      NS      c.ns.facebook.com.
facebook.com.   172800  IN      NS      d.ns.facebook.com.
;; Received 273 bytes from 192.35.51.30(f.gtld-servers.net.) in 11 ms

;; WARNING: No response from 129.134.30.12(a.ns.facebook.com.)

;; WARNING: No response from 185.89.219.12(d.ns.facebook.com.)

;; WARNING: No response from 129.134.31.12(b.ns.facebook.com.)

;; WARNING: No response from 185.89.218.12(c.ns.facebook.com.)

;; ERROR: No servers could be reached
```

Trace for Paris, France (Trial)

```
com.    172800  IN      NS      m.gtld-servers.net.
com.    172800  IN      NS      g.gtld-servers.net.
com.    172800  IN      NS      a.gtld-servers.net.
com.    172800  IN      NS      c.gtld-servers.net.
com.    172800  IN      NS      b.gtld-servers.net.
com.    172800  IN      NS      e.gtld-servers.net.
com.    172800  IN      NS      i.gtld-servers.net.
com.    172800  IN      NS      d.gtld-servers.net.
com.    172800  IN      NS      h.gtld-servers.net.
com.    172800  IN      NS      k.gtld-servers.net.
com.    172800  IN      NS      f.gtld-servers.net.
com.    172800  IN      NS      j.gtld-servers.net.
com.    172800  IN      NS      l.gtld-servers.net.
;; Received 490 bytes from 193.0.14.129(k.root-servers.net.) in 10 ms

facebook.com.   172800  IN      NS      a.ns.facebook.com.
facebook.com.   172800  IN      NS      b.ns.facebook.com.
facebook.com.   172800  IN      NS      c.ns.facebook.com.
facebook.com.   172800  IN      NS      d.ns.facebook.com.
;; Received 273 bytes from 192.35.51.30(f.gtld-servers.net.) in 11 ms

;; WARNING: No response from 129.134.30.12(a.ns.facebook.com.)

;; WARNING: No response from 185.89.219.12(d.ns.facebook.com.)

;; WARNING: No response from 129.134.31.12(b.ns.facebook.com.)

;; WARNING: No response from 185.89.218.12(c.ns.facebook.com.)

;; ERROR: No servers could be reached
```

Servidores DNS de Facebook

```
129.134.25.0/24
129.134.26.0/24
129.134.27.0/24
129.134.28.0/24
129.134.29.0/24
129.134.30.0/23
129.134.30.0/24
129.134.31.0/24
129.134.65.0/24
129.134.66.0/24
129.134.67.0/24
129.134.68.0/24
129.134.69.0/24
129.134.70.0/24
129.134.71.0/24
129.134.72.0/24
129.134.73.0/24
129.134.74.0/24
129.134.75.0/24
129.134.76.0/24
129.134.79.0/24
157.240.207.0/24
185.89.218.0/23
185.89.218.0/24
185.89.219.0/24
2a03:2880:f0fc::/47
2a03:2880:f0fc::/48
2a03:2880:f0fd::/48
2a03:2880:f0ff::/48
2a03:2880:f1fc::/47
2a03:2880:f1fc::/48
2a03:2880:f1fd::/48
2a03:2880:f1ff::/48
2a03:2880:f2ff::/48
2a03:2880:ff08::/48
2a03:2880:ff09::/48
2a03:2880:ff0a::/48
2a03:2880:ff0b::/48
2a03:2880:ff0c::/48
2a03:2881:4000::/48
2a03:2881:4001::/48
2a03:2881:4002::/48
2a03:2881:4004::/48
2a03:2881:4006::/48
2a03:2881:4007::/48
2a03:2881:4009::/48
69.171.250.0/24
```

Current Test
Facebook NIS
Settings
Server
129.134.30.12:53
Agent
All agents
Run Now
Save
Share

Views

DNS
DNS Server

NETWORK
Overview
Path Visualization

ROUTING
BGP Route Visualization

Metric
Loss

24h 7d 14d

Average Loss

100%
0%

18:30

Target Server
129.134.30.12:53

Showing data from **Mon, Oct 4 18:36 - 18:37 UTC** (6 minutes ago)

Latest

Node in Destination Network (with forwarding loss)

129.134.37.146

Warning: 1 trace terminates here

Prefix	129.134.0.0/17
Network	Facebook, Inc. (AS 32934)
Location	US
DSCP	Best Effort (DSCP 0)
Forwarding Loss	100% (6 of 6 packets)
Loss Frequency	Low
Avg. Response	148 ms
Affected Tests	100% of all tests 1 of your tests

Show only agents using this node

Path Visualization

8 hops
0 hops

Showing: 1 of 1 Test · 34 of 34 Agents · 4 of 4 Servers · Hide IP Address labels

Grouping: Agents by **Agent** · Interfaces by **Device** · Destinations by **No Grouping**

Highlighting: **Forwarding Loss > 25 %** (30 nodes) · **Link Delay > 20 ms** (27 links)

Selecting: Click a node or link **Info** (3)

Highlight nodes that match **all** / any

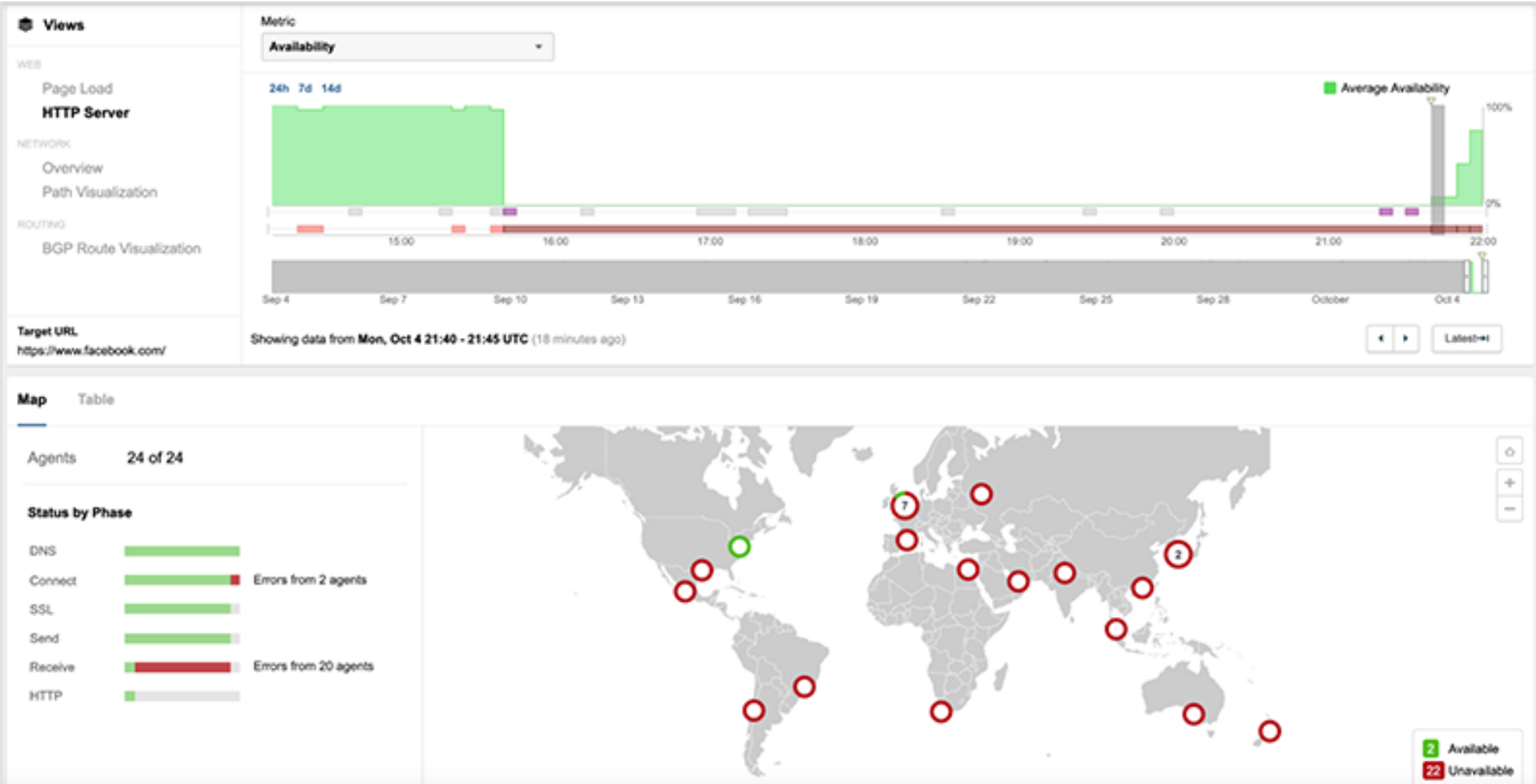
Country, IP address, Prefix, or Title...

Menu

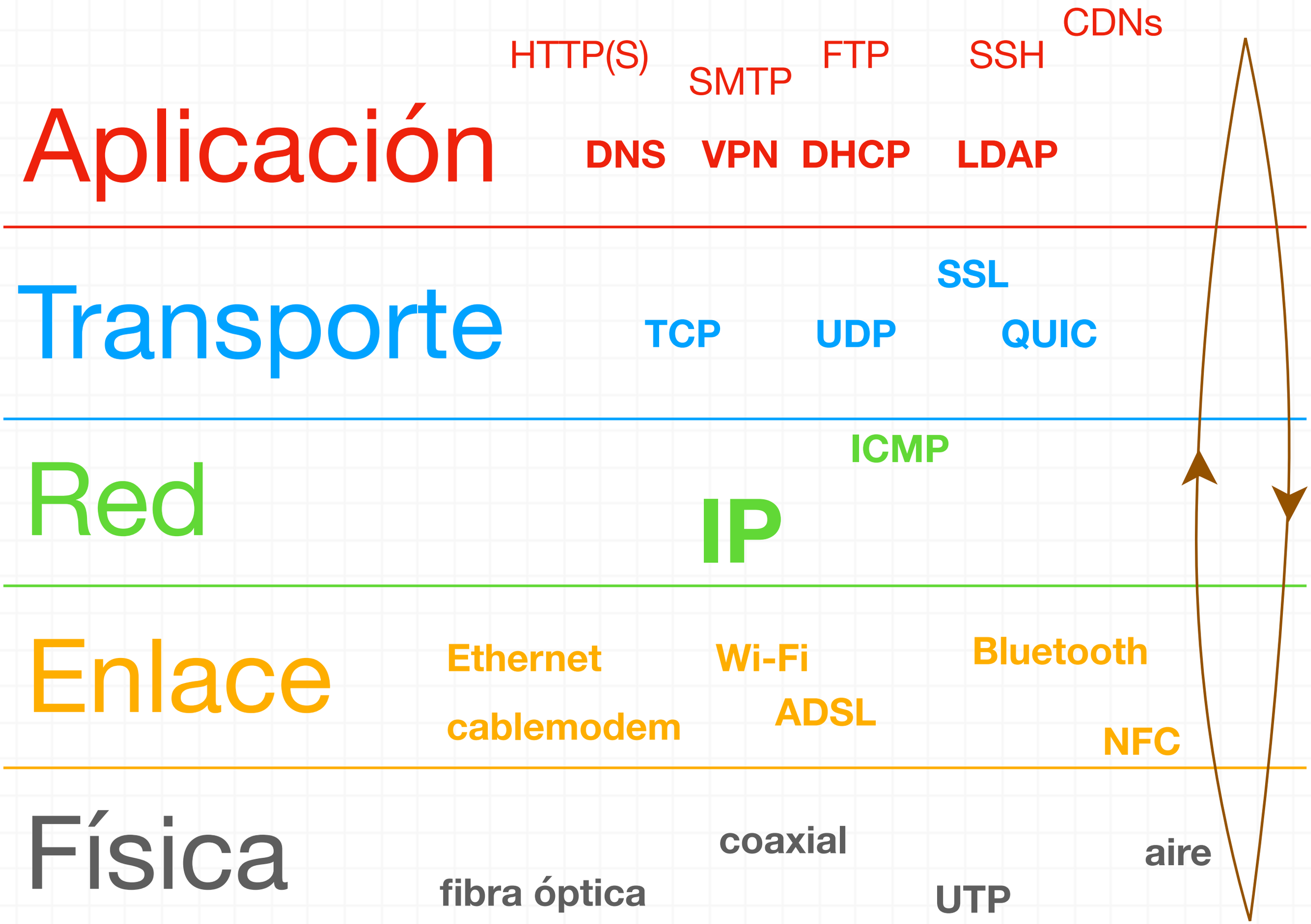
Buenos Aires, Argentina

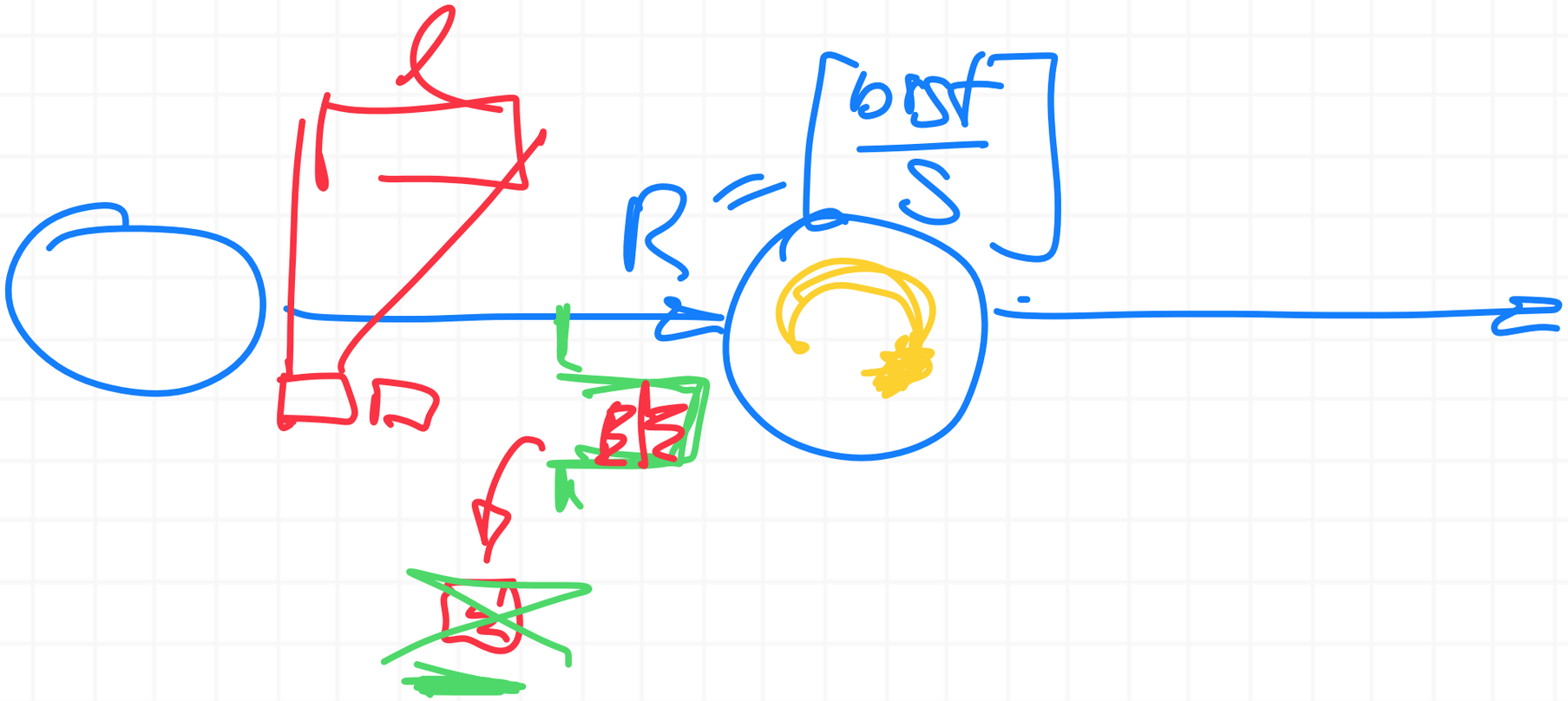
Brisbane, Australia

São Paulo, Brazil



Servidores no dieron abasto a responder.


Aplicación
HTTP(S)
SMTP
FTP
SSH
CDNs
DNS
VPN
DHCP
LDAP
Transporte
SSL
TCP
UDP
QUIC
Red
ICMP
IP
Enlace
Ethernet
Wi-Fi
Bluetooth
cablemodem
ADSL
NFC
Física
coaxial
aire
fibra óptica
UTP



T?

$$l = a \cdot b \ [\text{bits}]$$

$$t_{inserción} = \frac{l\,[\text{bits}]}{R\left[\frac{\text{bits}}{s}\right]} = [s] \qquad t_{transmisión} = \frac{D\,[m]}{\frac{2}{3}c\left[\frac{m}{s}\right]}$$

$$t_{buffer} = \frac{k \cdot l}{\lambda} \qquad t_{procesamiento}$$

$$T = \sum^{i} t_{inserción} + \sum^{i} t_{transmisión} + \sum^{j} t_{buffer} + \sum^{j} t_{proc.}$$

Las Aplicaciones son la razón de existir de Internet

Arquitecturas:

- Cliente-Servidor [datacenters]
- P2P

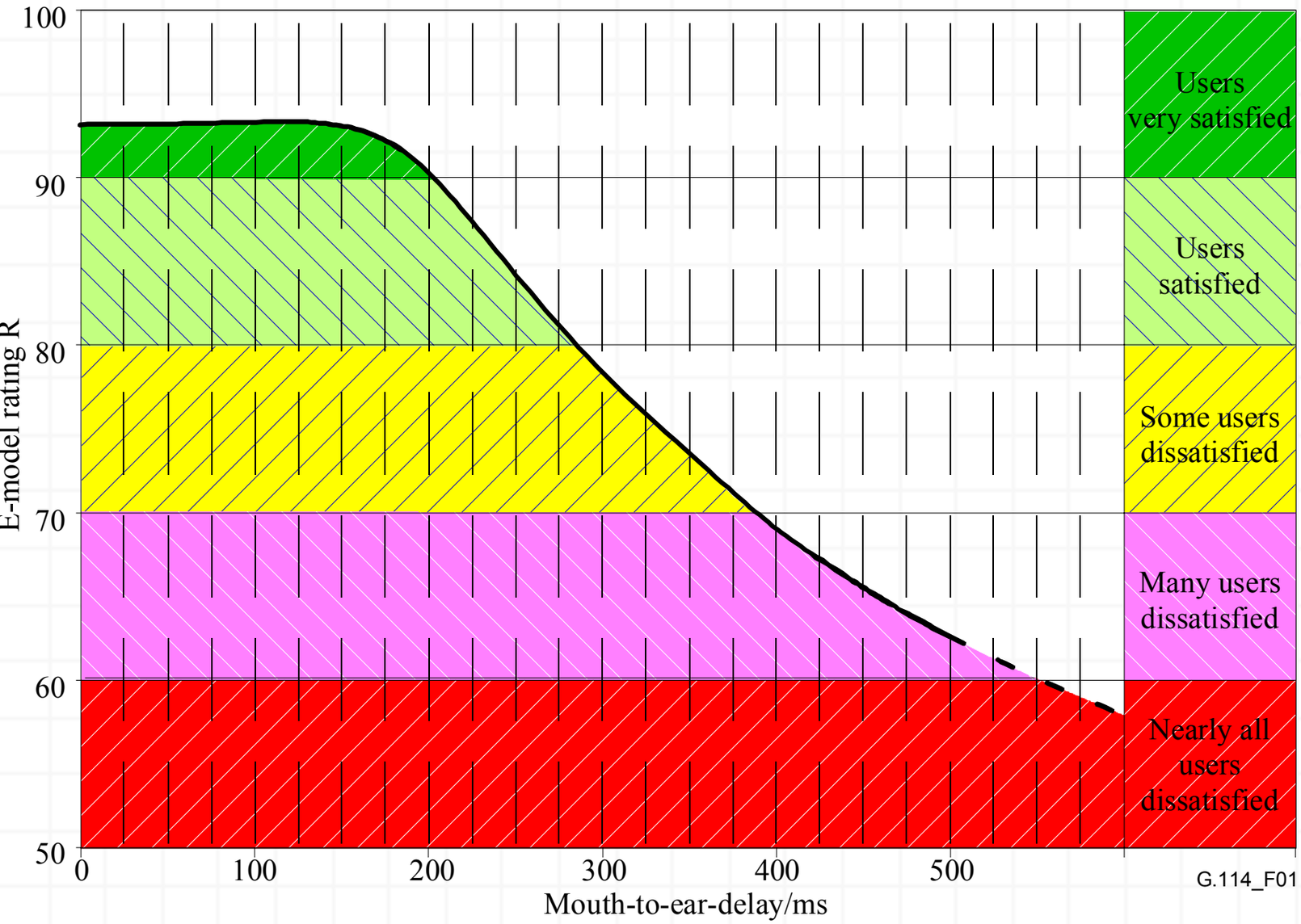
Comunicación entre procesos [mensajes, *socket()*, capa de transporte]

Características que ofrece el transporte:

- Transmisión confiable
- Caudal
- Sincronización
- Seguridad
- Con/sin pérdidas
- Conectado vs. desconectadas

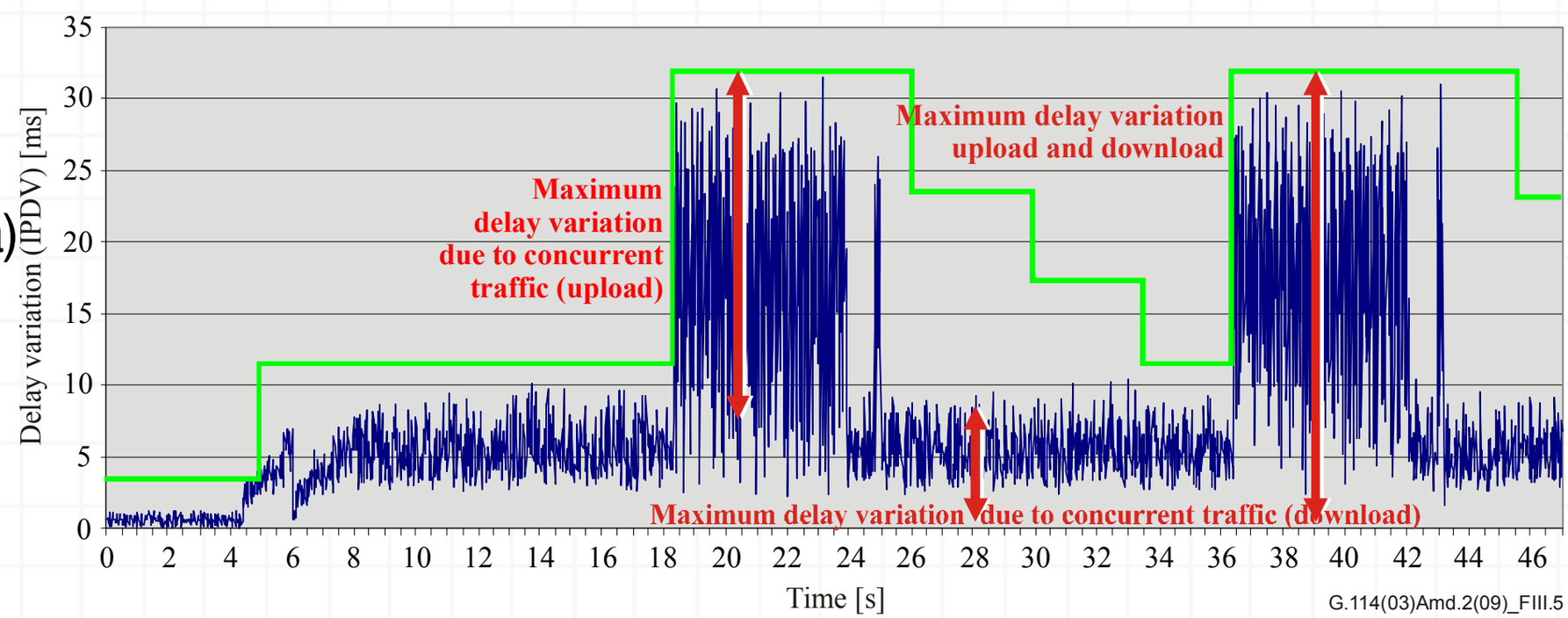
Paréntesis: Organismos Normalizadores

- IETF (Internet Engineer Task Force)
- ITU (International Telecommunication Union)
- W3C (World Wide Web Consortium)



Ejemplo: voz sobre IP (VoIP), en ITU-T Rec. G.114 (05/2003)

Ejemplo de jitter (variabilidad de la demora) en Rec. ITU-T G.114 (2003)/Amd.2 (11/2009)



Protocolo HTTP (HyperText Transfert Protocol)

- Aplicación Cliente-Servidor
- Permite transmitir texto formateado, imágenes, multimedia, etc.
- Conexiones: No-persistentes (1 conexión por elemento) vs. Persistentes (varios elementos por conexión)
- *Web Caching*

```
ihameli@aleph ~ % telnet 157.92.49.38 80
Trying 157.92.49.38...
Connected to www.fi.uba.ar.
Escape character is '^]'.
GET / HTTP/1.1
Host: www.fi.uba.ar
Connection: close
User-agent: Mozilla/5.0
Accept-language: en

HTTP/1.1 200 OK
Date: Tue, 14 Sep 2021 16:27:42 GMT
Server: Apache
X-Content-Type-Options: nosniff
Etag: "1631626276-0"
Content-Language: es
X-Frame-Options: SAMEORIGIN
X-UA-Compatible: IE=edge,chrome=1
Cache-Control: public, max-age=900
Last-Modified: Tue, 14 Sep 2021 13:31:16 GMT
Expires: Sun, 19 Nov 1978 05:00:00 GMT
Vary: Cookie,Accept-Encoding
Connection: close
Transfer-Encoding: chunked
Content-Type: text/html; charset=utf-8

24a0e
<!DOCTYPE html>
```

Wi-Fi: en0 (host 157.92.49.38)

Apply a display filter ... <⌘/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.0.7	157.92.49.38	TCP	78	63595 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=313275…
2	0.019424	157.92.49.38	192.168.0.7	TCP	74	80 → 63595 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PE…
3	0.019555	192.168.0.7	157.92.49.38	TCP	66	63595 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=3132756109 TSe…
4	5.225987	192.168.0.7	157.92.49.38	TCP	82	[TCP segment of a reassembled PDU]
5	5.246400	157.92.49.38	192.168.0.7	TCP	66	80 → 63595 [ACK] Seq=1 Ack=17 Win=29056 Len=0 TSval=1989624540 TSe…
6	5.246522	192.168.0.7	157.92.49.38	TCP	131	[TCP segment of a reassembled PDU]
7	5.262516	157.92.49.38	192.168.0.7	TCP	66	80 → 63595 [ACK] Seq=1 Ack=82 Win=29056 Len=0 TSval=1989624544 TSe…
8	6.039389	192.168.0.7	157.92.49.38	TCP	87	[TCP segment of a reassembled PDU]
9	6.056451	157.92.49.38	192.168.0.7	TCP	66	80 → 63595 [ACK] Seq=1 Ack=103 Win=29056 Len=0 TSval=1989624743 TS…
10	11.0383…	192.168.0.7	157.92.49.38	HTTP	68	GET / HTTP/1.1
11	11.0718…	157.92.49.38	192.168.0.7	TCP	66	80 → 63595 [ACK] Seq=1 Ack=105 Win=29056 Len=0 TSval=1989625993 TS…
12	11.0718…	157.92.49.38	192.168.0.7	TCP	1514	[TCP segment of a reassembled PDU]
13	11.0719…	192.168.0.7	157.92.49.38	TCP	66	63595 → 80 [ACK] Seq=105 Ack=1449 Win=130304 Len=0 TSval=313276713…
14	11.0721…	157.92.49.38	192.168.0.7	TCP	1514	[TCP Previous segment not captured] [TCP segment of a reassembled …
15	11.0722…	192.168.0.7	157.92.49.38	TCP	78	[TCP Window Update] 63595 → 80 [ACK] Seq=105 Ack=1449 Win=130624 L…

```
> Frame 10: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface 0
> Ethernet II, Src: 8c:85:90:0e:5e:fe (8c:85:90:0e:5e:fe), Dst: a8:6a:bb:cb:f1:d4 (a8:6a:bb:cb:f1:d4)
> Internet Protocol Version 4, Src: 192.168.0.7, Dst: 157.92.49.38
> Transmission Control Protocol, Src Port: 63595 (63595), Dst Port: 80 (80), Seq: 103, Ack: 1, Len: 2
> [4 Reassembled TCP Segments (104 bytes): #4(16), #6(65), #8(21), #10(2)]
v Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: www.fi.uba.ar\r\n
    Connection: close\r\n
    User-agent: Mozilla/5.0\r\n
    Accept-language: en\r\n
    \r\n
    [Full request URI: http://www.fi.uba.ar/]
    [HTTP request 1/1]
```

wireshark_pcapng_en0_20210914132733_vgGJGn

Packets: 218 · Displayed: 218 (100.0%) · Dropped: 0 (0.0%)

Profile: Default

DNS (Domain Name System)

- Relación: **NOMBRE ⟺ @IP**
- Base de datos ***jerárquica*** y ***distribuida*** (punto de falla, múltiples consultas, administración distribuida)
- Otros servicios:
    - Host aliasing
    - Mail server aliasing
    - Load Distribution
- Ejemplo de TLDs (Top Level Domains):

```
                                  |
                                  |
          +-------------------+-----------------+----------+-------+-...
          |                   |                 |          |       |
         MIL                 EDU               ARPA       AR      BR
          |                   |                 |          |       |
          |                   |                 |          |
     +-----+-----+            |     +------+-----+-----+    +----...
     |     |     |            |     |      |           |    |
    BRL  NOSC  DARPA          |  IN-ADDR  SRI-NIC     ACC  UBA
                              |                              |
     +--------+----------------+--------------+--------+    +----...
     |        |                |              |        |    |
    UCI      MIT               |            UDEL     YALE  FI
              |               ISI                          |
              |                |                           |
          +---+---+            |                           |
          |       |            |                           |
         LCS  ACHILLES  +--+-----+-----+--------+        +---+-----+...
          |             |  |     |     |        |        |   |     |
          XX            A  C   VAXA  VENERA Mockapetris  LABi WWW   CNET
```

Root servers en el mundo:

https://www.google.com/maps/d/viewer?mid=1LcHEpzl-7RzziWzDa4h3BxJcbEo&hl=en&usp=sharing

Tipos de consultas:

- Autorizadas o no
- Recursivas o iterativas
- Tipo:
    - A : nombre @IP
    - NS : Servidor DNS
    - CNAME: es el nombre canónico del host (puede tener muchos)
    - MX : Mail exchanger
    - SOA : *start of a zone of authority*
    - PTR : Pointer to Record

```
ihameli@aleph ~ % dig www.fi.uba.ar

; <<>> DiG 9.10.6 <<>> www.fi.uba.ar
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id:
15468
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0,
ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.fi.uba.ar.     INA

;; ANSWER SECTION:
www.fi.uba.ar.    3525 INA 157.92.49.38

;; Query time: 37 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Tue Sep 14 13:26:23 -03 2021
;; MSG SIZE  rcvd: 58
```

```
ihameli@aleph ~ % dig +trace  www.fi.uba.ar

; <<>> DiG 9.10.6 <<>> +trace www.fi.uba.ar
;; global options: +cmd
.        4351  IN NS m.root-servers.net.
.        4351  IN NS b.root-servers.net.
.        4351  IN NS c.root-servers.net.
```

Wi-Fi: en0 (port 53)

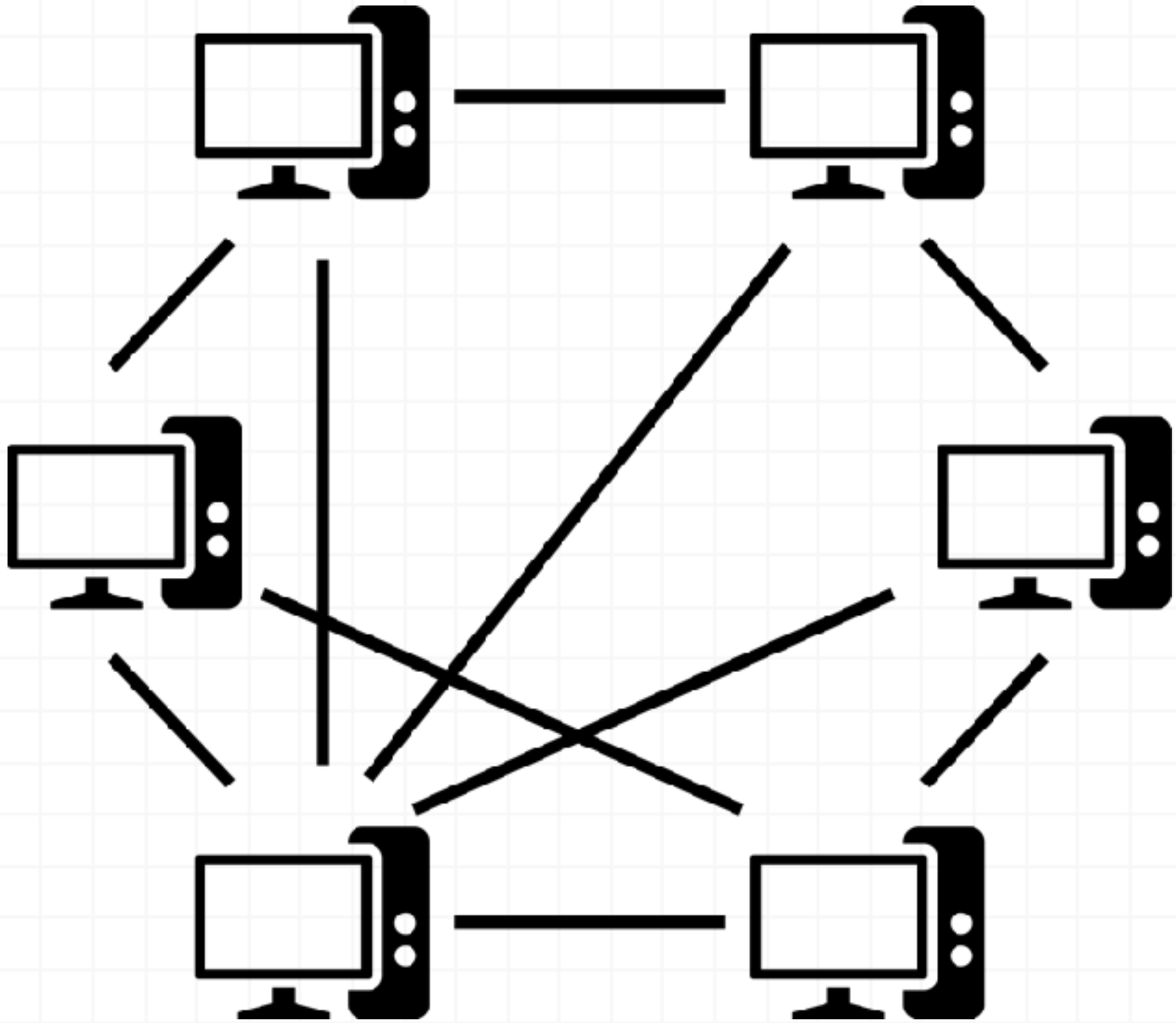
Apply a display filter ... <⌘/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.0.7	8.8.8.8	DNS	70	Standard query 0x3e0c NS <Root> OPT
2	0.018767	8.8.8.8	192.168.0.7	DNS	567	Standard query response 0x3e0c NS <Root> NS m.root-servers.net NS b.root-servers.net…
3	0.022615	192.168.0.7	8.8.8.8	DNS	78	Standard query 0xe439 A m.root-servers.net
4	0.022773	192.168.0.7	8.8.8.8	DNS	78	Standard query 0xd04c AAAA m.root-servers.net

```
Domain Name System (query)
    [Response In: 2]
    Transaction ID: 0x3e0c
    Flags: 0x0020 Standard query
      0... .... .... .... = Response: Message is a query
      .000 0... .... .... = Opcode: Standard query (0)
      .... ..0. .... .... = Truncated: Message is not truncated
      .... ...0 .... .... = Recursion desired: Don't do query recursively
      .... .... .0.. .... = Z: reserved (0)
      .... .... ..1. .... = AD bit: Set
      .... .... ...0 .... = Non-authenticated data: Unacceptable
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
    Queries
      <Root>: type NS, class IN
        Name: <Root>
        [Name Length: 0]
        [Label Count: 0]
        Type: NS (authoritative Name Server) (2)
        Class: IN (0x0001)
    Additional records
      <Root>: type OPT
        Name: <Root>
        Type: OPT (41)
        UDP payload size: 4096
        Higher bits in extended RCODE: 0x00
        EDNS0 version: 0
        Z: 0x8000
          1... .... .... .... = DO bit: Accepts DNSSEC security RRs
          .000 0000 0000 0000 = Reserved: 0x0000
        Data length: 0
```

wireshark_pcapng_en0_20210914134307_xEujdO Packets: 172 · Displayed: 172 (100.0%) Profile: Default

Arquitectura P2P (peer to peer)



The 360 Degree, CC BY-SA 4.0, via Wikimedia Commons

Para hacer (no es obligatorio): usar el comando “`dig`" con las opciones iterativa, autorizada y verborrágica capturando por la pantalla de la terminal y también mediante "`Wireshark`". Subirlo al Campus un informe en formato PDF.

Resumen:

Capa de Aplicación:

- necesidades
- interfases
- modelos de comunicaciones

HTTP: *(realizar captura con Wireshark)*

- arquitectura
- métodos
- proxy + caching

DNS: *(realizar captura con Wireshark)*

- arquitectura
- TLS
- propiedades

Consideraciones de las arquitecturas P2P

Próxima clase 21/3/2025: Capítulo 3, hasta *3.4 Principles of Reliable Data Transfer* inclusive.

Presente